

A
MAJOR PROJECT
on
PATH2HIRE : AI-POWERED PERSONALIZED
CAREER ROADMAPS

Submitted to
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD

In partial fulfilment of the requirement for the award of the degree of
BACHELOR OF TECHNOLOGY

in
COMPUTER SCIENCE AND ENGINEERING(CYBER SECURITY)

By

THOTA SREENIDHI	227Y1A6248
AMANDU AKSHAYA	227Y1A6203
SHAIK YOUSUF BASHA	227Y1A6264

Under the Guidance of
Mr.K.Pranay Kumar, Assistant Professor



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(CYBER SECURITY)



MARRI LAXMAN REDDY
INSTITUTE OF TECHNOLOGY & MANAGEMENT

(AN AUTONOMOUS INSTITUTION)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)

NAAC Accredited Institution with 'A' Grade & Recognized Under Section 2(f) & 12(B) of the UGC act, 1956

March , 2026



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(CYBER SECURITY)

Date: March-2026

CERTIFICATE

This is to certify that the project work entitled “**PATH2HIRE :AI POWERED PERSONALIZED CAREER ROADMAPS**” work done by **THOTA SREENIDHI (227Y1A6248), AMANDU AKSHAYA(227Y1A6203) and SHAIK YOUSUF BASHA (227Y1A6264)** student(s) of Department of Computer Science and Engineering(CSC), is a record of bonafide work carried out by the members during a period from December, 2025 to March ,2026 under the supervision of **Mr .PRANAY KUMAR**. This project is done as a fulfilment of obtaining Bachelor of Technology Degree to be awarded by Jawaharlal Nehru Technological University Hyderabad, Hyderabad.

The matter embodied in this project report has not been submitted by us to any other university for the award of any other degree.

THOTA SREENIDIHI

AMANDU AKSHAYA

SHAIK YOUSUF BASHA

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Date:

(Mr.K.Pranay Kumar)

The Viva-Voce Examination of above student(s), has been held on.....

Head of the Department

Principal/Director

External Examiner

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DECLARATION

We hereby declare that the Major Project report entitled, **“PATH2HIRE :AI POWERED PERSONALIZED CAREER ROADMAPS”** is record of work duly completed by us at Marri Laxman Reddy Institute of Technology and Management under the guidance of Mr. K Pranay Kumar, Assistant Professor for the reward of B. Tech Degree, this project work has not been submitted to any other University/Institution.

THOTA SREENIDHI

227Y1A6248

AMANDU AKSHAYA

227Y1A6203

SHAIK YOUSUF BASHA

227Y1A6264

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface
UX	User Experience
UML	Unified Modeling Language
DB	Database
SQL	Structured Query Language
HTTP	Hypertext Transfer Protocol
DB	Data base
SQL	Structured Query Language

HTTPS	Hypertext Transfer Protocol Secure
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ABSTRACT

Interview preparation is a critical challenge faced by students and job seekers, especially when transitioning from academic learning to industry-oriented roles. Many candidates struggle due to the lack of structured guidance, unclear understanding of job requirements, and limited exposure to real interview environments. Traditional preparation methods often rely on generic materials that do not align with specific job roles, resulting in inefficient preparation and reduced confidence. To address these challenges, this project presents an intelligent, web-based interview preparation system that provides role-specific guidance, structured learning support, and realistic interview practice.

The system enables users to input their target job role and corresponding job description, which serves as the foundation for personalized preparation. By analyzing the job description, the system identifies required skills, important topics, and expected interview areas relevant to the selected role. Based on this analysis, a structured learning roadmap is generated to help users plan their preparation effectively within a limited time frame. This approach ensures focused learning and reduces confusion caused by unorganized study methods.

A key feature of the system is the voice-based mock interview module, which simulates real interview scenarios. Users can answer interview questions verbally, allowing them to practice communication skills and experience interview pressure similar to real-world conditions. The system converts spoken responses into text and evaluates them using keyword-based analysis to assess relevance and understanding. Detailed feedback and performance scores are provided for each response, enabling users to identify strengths and areas for improvement.

The application is designed using modern web technologies with a modular architecture that separates the user interface, processing logic, and backend services. This design ensures scalability, maintainability, and efficient performance. The user interface is clean, responsive, and intuitive, making the system accessible even to users with minimal technical knowledge. Real-time processing, efficient data handling, and robust error management contribute to a smooth and reliable user experience.

Overall, the project aims to bridge the gap between academic knowledge and industry expectations by combining intelligent analysis, structured preparation, and practical interview practice. The system serves as an effective tool for improving interview readiness, boosting confidence, and supporting students and job seekers in achieving better career outcomes.

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Path2Hire is an AI-powered interview preparation and career guidance platform developed to assist students and job seekers in preparing for job interviews in a structured and effective manner. The project focuses on bridging the gap between academic learning and industry expectations by providing role-specific guidance, personalized learning plans, and realistic interview practice.

The system allows users to input their target job role and the corresponding job description. Using intelligent analysis, the application extracts key technologies, important concepts, and probable interview questions relevant to the selected role. Based on this analysis, Path2Hire generates a structured learning roadmap that helps users prepare systematically within a defined time frame.

A major highlight of the project is the voice-based mock interview module, where users answer interview questions using their voice. The system converts speech into text, evaluates responses using keyword-based analysis, and provides detailed feedback along with performance scores. This feature helps users improve not only their technical knowledge but also their communication and confidence. Overall, Path2Hire is designed as a complete interview preparation solution that combines modern web technologies, AI-driven analysis, and user-centric design. The project demonstrates practical application of full-stack development, serverless architecture, and AI integration, making it highly relevant for real-world career preparation and academic evaluation.

1.2 PURPOSE OF THE PROJECT

The purpose of Path2Hire is to assist students and job seekers in preparing for interviews in a structured and intelligent manner. This platform is designed to simplify interview preparation by providing role-specific guidance based on real job requirements. The key purposes are:

- To analyze job descriptions and identify industry-relevant skills and technologies.
- To guide users with a personalized learning roadmap instead of generic preparation methods.
- To help candidates understand important concepts and expected interview questions for their target role.

- To offer a voice-based mock interview environment that simulates real interview scenarios.
- To enhance communication skills and confidence through spoken interview practice.
- To evaluate user performance using keyword-based analysis and detailed feedback.
- To reduce uncertainty in interview preparation by providing clear and focused learning direction.
- To bridge the gap between academic learning and professional industry expectations.

Overall, Path2Hire serves as an intelligent career preparation solution that combines modern web technologies and AI-driven analysis to improve interview readiness and employability.

1.3 MOTIVATION

With the rapid growth of the technology industry, the recruitment process has become more competitive and dynamic. Students and job seekers are expected not only to have technical knowledge but also to demonstrate strong communication skills and confidence during interviews. However, many candidates are unsure about how to prepare effectively and often follow generic resources that do not match specific job roles.

Another motivating factor is the lack of personalized preparation tools that adapt to individual career goals. Candidates usually prepare multiple topics without knowing which ones are truly important for their target position. The absence of structured learning plans and realistic interview simulations makes it difficult to assess readiness before attending actual interviews.

These challenges inspired the development of this system, which focuses on providing role-based guidance, structured preparation, and interview practice in a single solution. The major motivations are outlined below:

- Growing competition in campus placements and job interviews
- Need for personalized guidance based on specific job roles
- Inefficient preparation due to non-targeted study materials
- Lack of tools for self-assessment and performance measurement
- Insufficient focus on communication and interview confidence
- Requirement for a systematic approach to interview readiness
- Desire to support students in transitioning from academics to professional careers

1.4 SCOPE OF THE PROJECT

The scope of this system defines the boundaries and functionalities covered to support interview preparation and career readiness for students and job seekers. It focuses on providing structured guidance, practical interview practice, and performance evaluation within a single platform.

The scope includes the following aspects:

- Allows users to enter their target job role and corresponding job description for analysis.
- Analyzes job descriptions to identify required skills, important topics, and expected interview questions.
- Generates a structured and time-based learning roadmap for interview preparation.
- Provides a voice-based mock interview environment with multiple interview questions.
- Converts spoken answers into text for evaluation and review.
- Evaluates interview responses using keyword-based analysis and provides performance scores.
- Displays detailed feedback to help users identify strengths and areas for improvement.
- Supports preparation for both technical and behavioral interview questions.

The system is primarily intended for students, fresh graduates, and job seekers preparing for interviews. Future extensions can include user authentication, interview history tracking, advanced analytics, and integration with job portals.

1.5 OBJECTIVES

The main objective of this system is to help students and job seekers prepare effectively for job interviews by providing structured guidance, practical interview experience, and performance evaluation. The specific objectives are as follows:

- To analyze job descriptions and identify role-specific skills and technologies.
- To provide clear guidance on important topics required for interview preparation.
- To generate a personalized learning roadmap for systematic preparation.
- To simulate real interview scenarios through a voice-based mock interview process.
- To improve communication skills and interview confidence of users.
- To evaluate interview responses using keyword-based analysis.
- To provide detailed feedback and performance scores for self-improvement.
- To bridge the gap between academic knowledge and industry requirements.

1.6 LIMITATIONS

Despite providing effective interview preparation support, the system has certain limitations that should be considered:

- The accuracy of job analysis depends on the quality and completeness of the job description provided by the user.
- Keyword-based evaluation may not fully capture the depth or context of an answer.
- The system does not currently support long-term storage of user progress or interview history.
- Performance evaluation focuses on predefined keywords and may not account for creative or alternative valid responses.
- Internet connectivity is required for proper functioning of analysis and interview features.
- Background noise may affect the accuracy of voice-to-text transcription.
- The platform is designed for preparation purposes and does not guarantee interview success.

These limitations highlight areas where the system can be improved and extended in future versions.

CHAPTER 2

LITERATURE SURVEY

2.1 PREFERRED LANGUAGE

The system is developed primarily using JavaScript with TypeScript support, which serves as the main programming language for the application. This language is used extensively in frontend development and plays a central role in implementing the overall functionality of the system.

JavaScript is a widely used, high-level programming language that enables dynamic and interactive behavior in web applications. It is responsible for handling user interactions, form submissions, navigation between pages, and real-time updates within the application. Its event-driven and asynchronous nature makes it well suited for tasks such as API communication, audio recording, and data processing.

TypeScript is used as an extension of JavaScript to enhance code reliability and maintainability. It introduces static typing, which helps in identifying errors during development rather than at runtime. This improves code quality, makes debugging easier, and ensures better readability.

For frontend development, the system uses JavaScript and TypeScript in combination with the React library. React allows the interface to be built using reusable components, making the user interface modular and easy to maintain. It uses a virtual DOM mechanism that improves performance by updating only the necessary parts of the interface when changes occur. This results in a smooth and responsive user experience.

The preferred programming language also integrates well with modern tools and frameworks such as build systems, styling libraries, and routing mechanisms. It supports rapid development, easy testing, and scalability. Due to its wide adoption, strong community support, and compatibility with modern web standards, JavaScript with TypeScript is an ideal choice for both academic projects and real-world applications.

2.2 EXISTING SYSYEM

In the existing interview preparation methods, students and job seekers mostly rely on traditional learning approaches such as textbooks, online articles, video tutorials, and generic interview preparation websites. These resources provide general information but are not tailored to specific job roles or individual requirements.

Most existing platforms do not analyze real job descriptions. As a result, candidates often prepare unnecessary topics while missing important role-specific skills. Preparation is usually unstructured, and users must decide on their own what to study and in what order, which can be confusing and time-consuming.

Interview practice in the existing system is limited. Many platforms focus only on theoretical questions or written practice and do not provide realistic interview simulations. Voice-based interview practice and real-time feedback are often missing, which affects confidence and communication skills.

Additionally, existing systems rarely offer detailed performance evaluation. Candidates do not receive clear feedback on their answers or insights into areas that need improvement. Due to these limitations, there is a significant gap between academic preparation and actual industry interview expectations.

2.3 PROPOSED SYSTEM

The proposed system is an intelligent and user-friendly interview preparation solution designed to overcome the limitations of existing methods. It focuses on providing structured, role-specific guidance and realistic interview practice to help students and job seekers prepare more effectively.

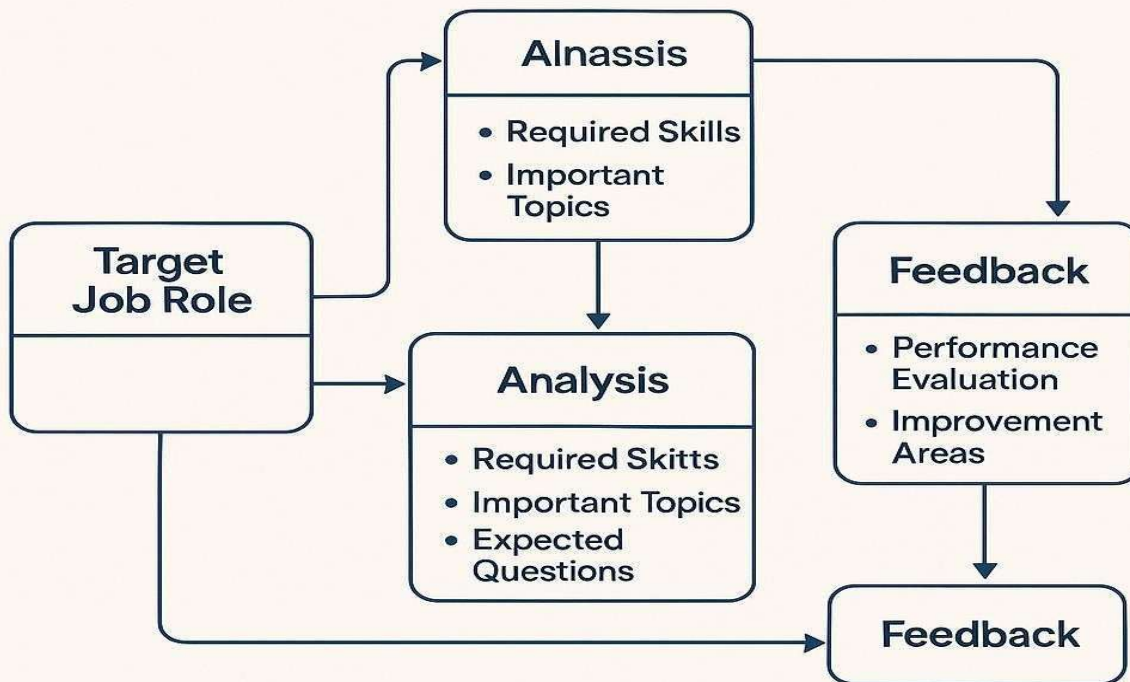
The system allows users to input their target job role and corresponding job description. Based on this input, it analyzes industry requirements and extracts relevant skills, important topics, and expected interview questions. This helps users clearly understand what is required for their chosen role.

A key feature of the proposed system is the personalized learning roadmap, which guides users on what to study and how to prepare within a specific time frame. This structured approach reduces confusion and improves preparation efficiency.

The system also includes a voice-based mock interview module that simulates real interview conditions. Users can answer questions verbally, receive automated evaluation, and view detailed feedback on their performance. This improves communication skills, confidence, and interview readiness.

Overall, the proposed system provides a complete and integrated interview preparation environment by combining intelligent analysis, structured learning, and practical interview practice, making it more effective than traditional preparation methods.

Proposed System



2.3.1 User workflow

The user workflow diagram shows the step-by-step flow from entering job details to receiving interview feedback.

CHAPTER 3

SYSTEM ANALYSIS

3.1 SYSTEM REQUIREMENTS

The system requirements define both the hardware and software resources needed for the proper development and execution of the application. Suitable hardware ensures smooth performance and supports features such as audio recording, while the software environment enables efficient processing, scalability, and reliable user interaction using modern web technologies.

3.2.1 : Software Requirements

Component	Specification
Operating System	Windows 10 or higher / Linux / macOS
Programming Language	JavaScript with TypeScript
Frontend Framework	React
Build Tool	Vite
Styling Framework	Tailwind CSS
UI Component Library	Component-based UI library
IDE / Code Editor	VS Code
Database	PostgreSQL
Version Control	Git / GitHub
API Communication	HTTP/REST APIs

Table 3.2.1: Software Requirements of the Proposed System

3.2.2 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent and above
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 20 GB HDD/SSD
Microphone	Required for voice interviews
Network	Broadband Internet connection

Table 3.2.2: Hardware Requirements of the Proposed System

3.2 FEASIBILITY STUDY

The feasibility study determines whether the proposed system can be practically developed and successfully implemented. The feasibility analysis is carried out in five major dimensions: technical, economic, operational, legal, and schedule feasibility.

1. Technical Feasibility

Technical feasibility examines whether the technology, tools, and resources required to build the system are available and suitable.

Availability of Technology

The system is developed using modern, reliable, and well-supported web technologies:

- JavaScript and TypeScript for application logic
- React framework for frontend development
- Vite as a fast build and development tool
- Tailwind CSS for responsive user interface design
- Serverless backend functions for processing requests
- PostgreSQL for structured data management
- Browser-based APIs for audio recording and processing

All these technologies are widely used in the industry, easy to learn, and supported by strong community documentation.

- A standard desktop or laptop computer

- Built-in or external microphone for voice-based interviews
- Minimum 4 GB RAM for smooth execution
- Processor: Intel i3 or equivalent and above
- Stable internet connection

The system does not require advanced hardware such as GPUs or high-end servers.

Performance and Compatibility

- Works on common operating systems such as Windows, Linux, and macOS
- Compatible with modern web browsers
- Supports smooth audio recording and real-time interaction

Feasibility Conclusion

The availability of required technologies and basic hardware makes the system technically feasible.

3. Economic Feasibility

Economic feasibility determines whether the system is affordable to develop and operate.

Development Cost

The development cost is minimal because:

- Most frameworks and libraries used are open-source
- No paid software licenses are required
- Development tools are freely available
- No need for dedicated physical servers
- Minimal storage requirements
- Efficient serverless execution reduces infrastructure cost

Maintenance Cost

- Occasional updates to improve system performance
- Minor changes for feature enhancement
- No continuous high maintenance effort required

Feasibility Conclusion

The system is economically feasible as it requires low development, operational, and maintenance costs.

4. Legal Feasibility

Legal feasibility checks whether the system follows legal rules, software licenses, and data privacy policies.

- All technologies used are open-source

- Tools and frameworks are permitted for academic and educational use
- The system does not permanently store personal data
- Audio responses are processed temporarily
- No sensitive user information is collected
- Follows standard data protection and privacy practices
- Does not violate software usage or copyright laws

Feasibility Conclusion

There are no legal constraints affecting the system, making it legally feasible.

5. Schedule Feasibility

Schedule feasibility determines whether the system can be completed within the planned time period.

- Use of modern frameworks speeds up development
- Modular architecture allows step-by-step implementation
- Development can be divided into frontend, backend, and testing phases
- The project fits well within the academic semester timeline

Feasibility Conclusion

The system can be successfully completed within the allotted time, making it schedule feasible.

3.3 DATA SETS USED IN SYSTEM

The system uses structured and unstructured data inputs to support analysis, interview preparation, and performance evaluation. Instead of relying on large pre-collected datasets, the application works primarily with real-time and user-provided data, making it flexible and role-specific.

The main data sets used in the system are described below:.

- **Job Role Data**
The job role entered by the user is used as a reference dataset to customize learning roadmaps and generate interview questions specific to that role.
- **Interview Question Dataset**
A dynamically generated set of technical and behavioral interview questions is used during mock interviews. Each question is associated with a predefined set of keywords for evaluation.
- **Keyword Dataset**
A keyword dataset is used to evaluate user responses. These keywords represent expected concepts and skills that should appear in an ideal answer.
- **User Response Data**
User responses, converted from voice to text, act as temporary datasets for performance analysis and feedback generation. These responses are processed during the session and are not permanently stored.

3.4 ARCHITECTURE

The system architecture follows a modern web-based, modular design that separates the user interface, application logic, and backend processing to ensure scalability, maintainability, and efficient performance. The architecture is designed to support real-time interaction, intelligent analysis, and voice-based interview processing.

At the client side, the system uses a web browser as the primary interface. Users interact with the application through a responsive frontend built using modern web technologies. This layer handles user inputs such as job role, job description, navigation through learning roadmaps, and voice-based interview responses. The browser also manages audio recording using built-in media APIs and sends the captured data to the backend for processing.

The frontend communicates with the backend through secure HTTP-based APIs. All requests related to job analysis, interview question generation, and voice transcription are sent to serverless backend services. This approach removes the need for traditional server management and allows the system to scale automatically based on demand.

The backend layer consists of serverless processing units that handle different responsibilities. One unit analyzes job descriptions to extract required skills, topics, and interview questions. Another unit generates interview questions along with evaluation keywords. A separate processing unit handles voice-to-text conversion by processing audio input received from the frontend. Each unit operates independently, improving system reliability and performance.

The database layer supports structured data management and future enhancements. While the current system processes data in real time without long-term storage, the architecture allows integration of user authentication, interview history storage, and progress tracking when required. This makes the system extensible and future-ready.

Overall, the architecture ensures clear separation of concerns, efficient data flow, real-time processing, and a smooth user experience. The modular design makes the system easy to maintain, upgrade, and deploy in both academic and real-world environments.

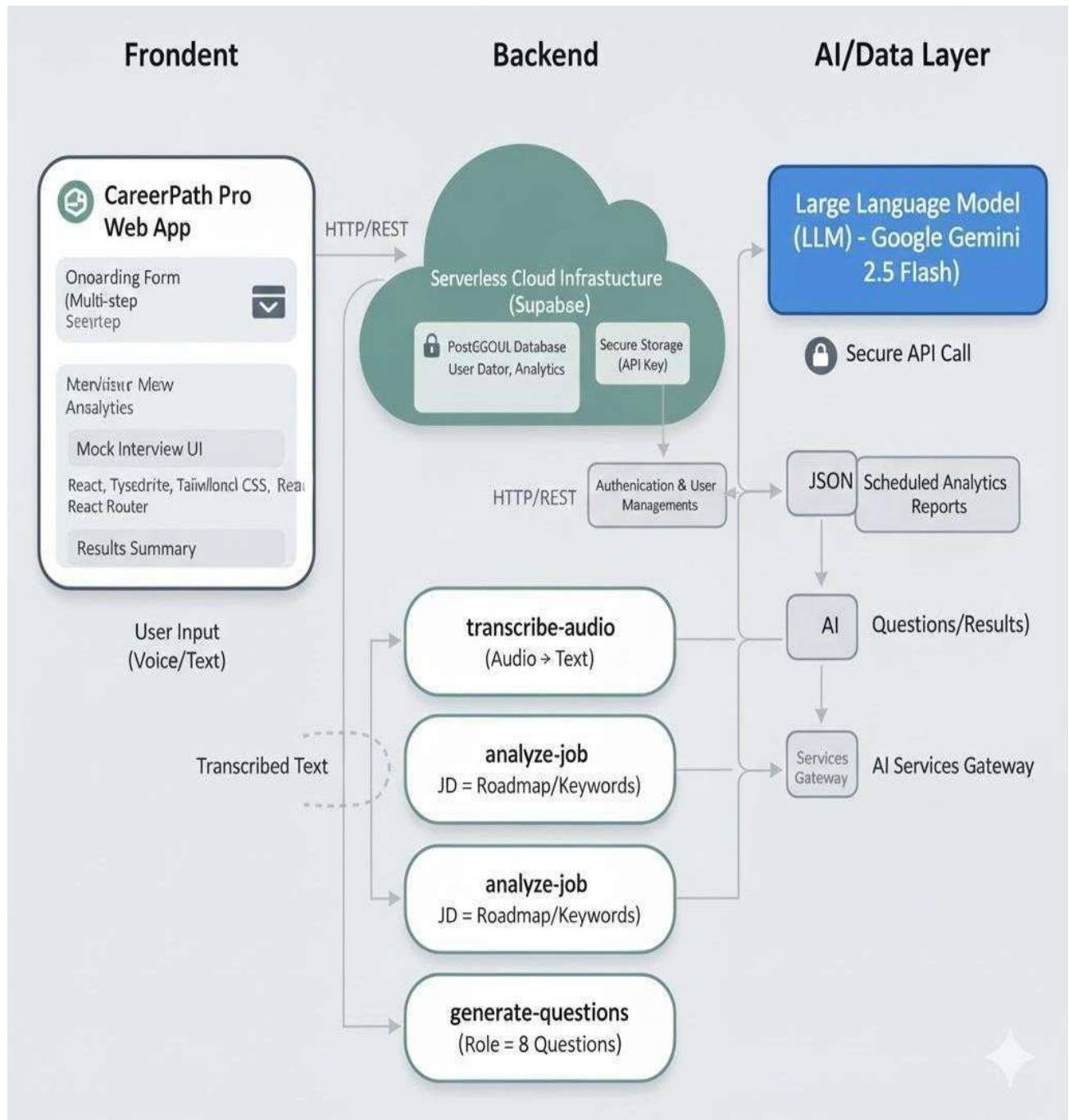


Fig 3.4: system architecture diagram

The system architecture diagram shows how different modules interact to process user input, perform analysis, and generate interview feedback. It explains the overall flow between frontend, backend, and evaluation components.

3.5 COMPONENTS /USER IN THE SYSTEM

The system includes several components and users that work together to deliver an effective and seamless interview preparation experience. Each component plays an important role in ensuring proper data flow, interaction, and processing within the application.

The primary user of the system is a student, job seeker, or fresh graduate who wants to prepare for job interviews. The user interacts with the application through a web browser to enter the target job role and job description, follow the learning roadmap, participate in voice-based mock interviews, and view feedback and performance results.

The web browser serves as the main interface between the user and the system. It is responsible for rendering the user interface, capturing user inputs, and recording audio using built-in browser APIs. The browser sends user data and audio responses to the backend and displays the processed results returned by the system.

The frontend application handles all user interactions and presentation logic. It manages form inputs, navigation, and the display of job analysis results, interview questions, and feedback. This component ensures that the system is responsive, interactive, and easy to use across different devices.

The API communication layer enables secure and efficient data exchange between the frontend and backend components. It transfers job descriptions, interview requests, and audio data for processing and receives structured responses from backend services.

The backend processing components perform the core operations of the system. These include analyzing job descriptions, generating interview questions, converting voice input into text, and evaluating interview responses. Each backend component operates independently to improve reliability and scalability.

The database component supports structured data handling and future enhancements. Although the current system processes most data in real time, the database allows for possible extensions such as storing user details, interview history, and performance analytics. Together, these components form a complete system that provides structured guidance, realistic interview practice, and meaningful performance evaluation.

CHAPTER 4

SYSTEM DESIGN

4.1 DESIGN OBJECTIVE

The design objectives define the goals and principles followed while designing the system to ensure efficiency, usability, and scalability. The major design objectives of the system are as follows:

- To design a user-friendly and intuitive interface that allows users to navigate the system easily without technical knowledge.
- To ensure a modular and scalable system design that supports future enhancements and feature expansion.
- To provide a responsive design that works efficiently across different devices and screen sizes.
- To enable real-time processing for job analysis, mock interviews, and performance evaluation.
- To maintain clear separation between user interface, processing logic, and data handling components.
- To ensure fast response time and smooth interaction throughout the system.
- To protect user data and maintain privacy during all system operations.
- To design the system in a way that minimizes errors and improves reliability.
- To support easy maintenance and updates with minimal changes to existing components.

4.2 INTERFACE DESIGN

The interface design of the system is created with a strong emphasis on usability, clarity, and consistency to ensure a smooth experience for all users. The overall layout follows a step-by-step structure that guides users logically through different stages such as onboarding, job description input, analysis results, learning roadmap, mock interviews, and performance feedback. This structured flow helps users understand the process easily and reduces confusion during interaction. The design follows a minimalistic approach, avoiding unnecessary visual clutter and focusing on essential elements. Proper alignment, spacing, and contrast are used to improve readability and visual balance. Clear headings, labels, and instructions are provided at every stage so that users can complete tasks without external assistance. The interface maintains consistency in design elements such as buttons, fonts, and color themes across all pages.

Responsiveness is an important aspect of the interface design. The system adapts smoothly to different screen sizes, ensuring proper usability on desktops, laptops, and tablets. Interactive components such as forms, tabs, cards, and progress indicators help present information in an organized and user-friendly manner. Visual cues such as icons and indicators assist users in understanding actions like recording audio, submitting responses, or navigating between sections. The interface also focuses on providing immediate feedback to user actions. Loading indicators, success messages, and error notifications inform users about system status and ongoing processes. The mock interview interface is designed to be clear and distraction-free, allowing users to focus on answering questions confidently. Overall, the interface design supports an engaging, professional, and efficient environment that enhances user experience and supports effective interview preparation.

4.3 UML DIAGRAMS

UML (Unified Modeling Language) diagrams are used in the system to visually represent the structure, behavior, and interaction of different components. These diagrams help in understanding how the system is designed, how data flows between components, and how users interact with various functionalities. UML diagrams provide a clear and standardized way to document the system design, making it easier to analyze, implement, and maintain.

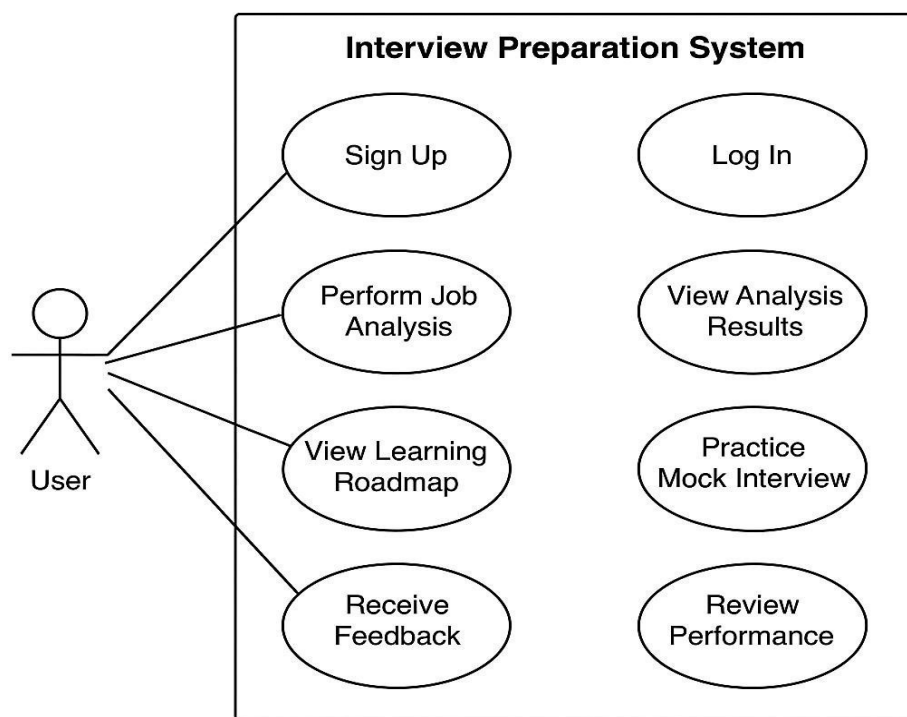
In this project, UML diagrams are used to represent both static and dynamic aspects of the system. They illustrate user interactions, system processes, and relationships between different components. By using UML diagrams, complex system logic is simplified into visual models, which improves clarity and communication during development and evaluation.

The UML diagrams also help in identifying system requirements, understanding module dependencies, and validating system behavior. Overall, the use of UML diagrams enhances system documentation and provides a clear blueprint of the system's architecture and functionality, making it easier for developers, reviewers, and evaluators to understand the complete working of the system.

4.3.1 Use case Diagram

The use case diagram represents the functional behavior of the system and illustrates how users interact with the application to achieve specific goals. It provides a high-level view of the system's capabilities and clearly defines the relationship between the user and various system functionalities.

In this system, the primary actor is the user, who represents students, job seekers, or fresh graduates preparing for interviews. The user interacts with the system through a web interface to perform different actions related to interview preparation.



4.3.1 Use case diagram

4.3.2 Sequence Diagram

The sequence diagram illustrates the step-by-step interaction between the user, frontend interface, backend services, and supporting components during the execution of system functionalities. It focuses on the order in which messages are exchanged and operations are performed to complete a user request. In the system, the sequence begins when the user interacts with the application through the web interface by entering job details or starting a mock interview. The frontend processes the input and sends requests to the backend services. The backend then performs tasks such as job description

analysis, interview question generation, or voice-to-text conversion and returns the processed results to the frontend.

The sequence of operations describes the step-by-step flow of activities performed by the system from user interaction to result generation.

1. The user accesses the system through a web browser and enters the required details such as job role and job description.
2. The frontend application validates the input and sends the job details to the backend services.
3. The backend analyzes the job description and extracts required skills, important topics, and expected interview questions.
4. The processed analysis results are sent back to the frontend and displayed to the user.
5. The system generates a personalized learning roadmap based on the analysis.
6. The user initiates the mock interview process through the interface.
7. Interview questions are presented one by one to the user.
8. The user answers each question using voice input.
9. The system captures the audio, converts it into text, and evaluates the response.
10. Performance data and feedback are generated for each answer.
11. After completing the interview, an overall performance summary is displayed to the user.
12. The session ends after the user reviews the feedback and results.

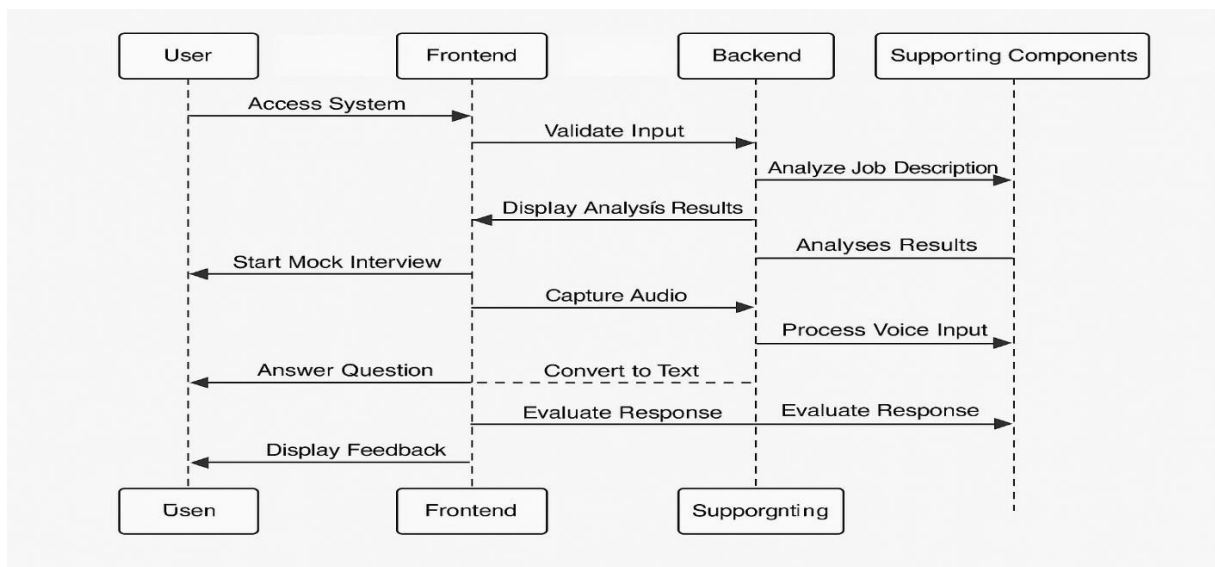


Fig 4.3.2 Sequence Diagram

4.4 MODULE DIAGRAM

The modular diagram represents the high-level organization of the system and shows how different functional modules interact with each other to provide a complete interview preparation solution. The system is divided into well-defined modules to ensure clarity, maintainability, and scalability.

The **User Module** represents the interaction point for students and job seekers. It manages user inputs such as job role selection, job description submission, interview participation, and viewing of feedback. This module ensures a smooth and guided user experience.

The **Frontend Interface** acts as the presentation layer of the system. It displays analysis results, learning roadmaps, interview questions, and performance feedback. It also handles user navigation and captures voice input during mock interviews.

The **Backend Processing Module** functions as the core of the system. It receives requests from the frontend, coordinates communication between different modules, and manages overall system logic. This module ensures secure and efficient data processing.

The **Job Description Analyzer Module** processes the job description provided by the user. It extracts important skills, technologies, and topics required for the target role. This information is essential for generating relevant interview questions and learning roadmaps.

The **Roadmap Generator Module** creates a structured and time-based learning plan based on the job analysis. It guides users on what topics to study and how to prepare systematically.

The **Mock Interview Module** manages the interview simulation process. It presents interview questions, controls interview flow, and collects user responses using voice input.

The **Voice-to-Text Module** converts spoken answers into text form, enabling automated evaluation of interview responses.

The **Evaluation Module** analyzes user answers using keyword-based scoring techniques. It measures relevance, identifies strengths and weaknesses, and generates performance scores.

The **Feedback Module** presents detailed feedback and results to the user, helping them understand their performance and improve future interview attempts.

Overall, the modular design ensures efficient interaction between components, supports easy system maintenance, and allows future enhancements such as performance tracking and advanced analytics.

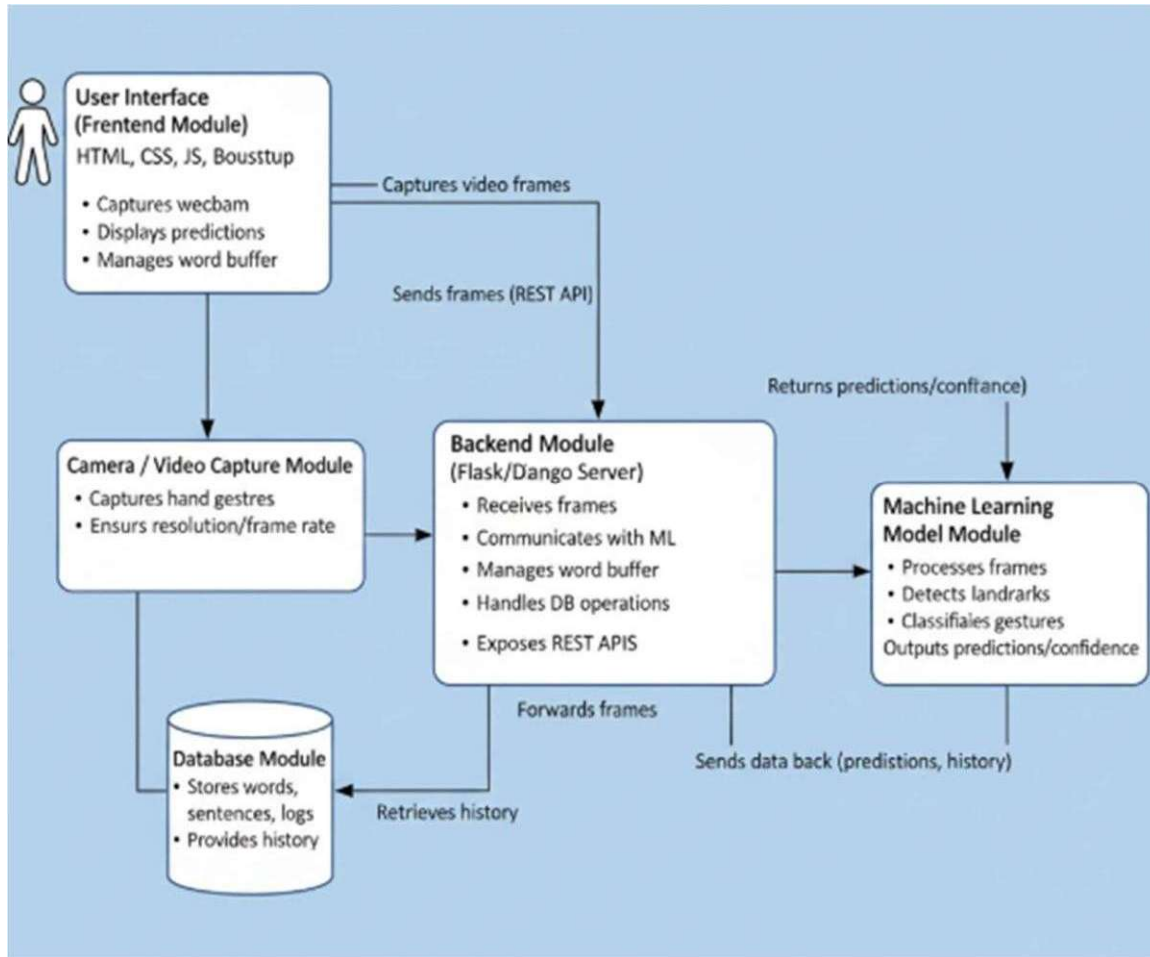


Fig 4.4 Modular Diagram

The modular diagram shows the major functional modules of the system and their interactions. It explains how each module contributes to job analysis, interview processing, evaluation, and feedback generation.

CHAPTER 5

IMPLEMENTATION

5.1 CODE SNIPPETS

5.1.1 Package-lock.json

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To8D2GPA==",
387     "license": "MIT",
388     "dependencies": {
389       "@radix-ui/primitive": "1.1.3",
390       "@radix-ui/react-collapsible": "1.1.12",
391       "@radix-ui/react-collection": "1.1.7",
392       "@radix-ui/react-compose-refs": "1.1.2",
393       "@radix-ui/react-context": "1.1.2",
394       "@radix-ui/react-direction": "1.1.1",
395       "@radix-ui/react-id": "1.1.1",
396       "@radix-ui/react-primitive": "2.1.3",

```

```

397     "@radix-ui/react-use-controllable-state": "1.2.2"
398   },
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403     "react-dom": "^16.8 || ^17.0 || ^18.0 || ^19.0 || ^19.0.0-rc"
404   },
405   "peerDependenciesMeta": {
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407       "optional": true
408     },
409     "@types/react-dom": {
410       "optional": true
411     }
412   }
413 },
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416   "resolved": "https://registry.npmjs.org/@radix-ui/react-alert-dialog/-
/react-alert-dialog-1.1.15.tgz",
417   "integrity": "sha512-
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425     "@radix-ui/react-slot": "1.2.3"
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431     "react-dom": "^16.8 || ^17.0 || ^18.0 || ^19.0 || ^19.0.0-rc"
432   },
433   "peerDependenciesMeta": {
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436     },
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439     }
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```

```

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449     },
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467     },
468     "peerDependencies": {
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472       "react-dom": "^16.8 || ^17.0 || ^18.0 || ^19.0 || ^19.0.0-rc"
473     },
474     "peerDependenciesMeta": {
475       "@types/react": {
476         "optional": true
477       },
478       "@types/react-dom": {
479         "optional": true
480       }

```

```

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496       "react-dom": "^16.8 || ^17.0 || ^18.0 || ^19.0 || ^19.0.0-rc"
497     },
498     "peerDependenciesMeta": {
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503         "optional": true
504       }
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519         "react-dom": "^16.8 || ^17.0 || ^18.0 || ^19.0 || ^19.0.0-rc"
520       },
521       "peerDependenciesMeta": {

```

```

521     "@types/react": {
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533   "license": "MIT",
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537     "@radix-ui/react-use-callback-ref": "1.1.1",
538     "@radix-ui/react-use-is-hydrated": "0.1.0",
539     "@radix-ui/react-use-layout-effect": "1.1.1"
540   },
541   "peerDependencies": {
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543     "@types/react-dom": "*",
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545     "react-dom": "^16.8 || ^17.0 || ^18.0 || ^19.0 || ^19.0.0-rc"
546   },
547   "peerDependenciesMeta": {
548     "@types/react": {
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550     },
551     "@types/react-dom": {
552       "optional": true
553     }
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558   "resolved": "https://registry.npmjs.org/@radix-ui/react-context/-/react-
context-1.1.3.tgz",
559   "integrity": "sha512-
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```

```

561     "peerDependencies": {
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564     },
565     "peerDependenciesMeta": {
566       "@types/react": {
567         "optional": true
568       }
569     }
570   },

```

5.1.2 index.html

```

<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>CareerPath Pro - AI-Powered Interview Preparation</title>
    <meta name="description" content="Land your dream job with AI-powered job analysis,
personalized learning roadmaps, and voice-based mock interviews. Prepare for your next interview
with confidence." />
    <meta name="author" content="CareerPath Pro" />
    <meta name="keywords" content="career growth, interview preparation, mock interview, job
analysis, AI career coach" />

    <meta property="og:title" content="CareerPath Pro - AI-Powered Interview Preparation" />
    <meta property="og:description" content="Get personalized job analysis, curated learning
roadmaps, and practice with AI-powered mock interviews." />
    <meta property="og:type" content="website" />
    <meta property="og:image" content="https://lovable.dev/opengraph-image-p98pqq.png" />

    <meta name="twitter:card" content="summary_large_image" />
    <meta name="twitter:title" content="CareerPath Pro - Land Your Dream Job" />
    <meta name="twitter:description" content="AI-powered interview preparation platform for
students and job seekers." />

```

```

<meta name="twitter:image" content="https://lovable.dev/opengraph-image-p98pqq.png" />

<link rel="canonical" href="/" />
</head>

<body>
  <div id="root"></div>
  <script type="module" src="/src/main.tsx"></script>
</body>
</html>

```

5.1.3 Nexttheme.js

```

"use client";
import {
  require_react
} from "./chunk-BG45W2ER.js";
import {
  __toESM
} from "./chunk-HXA6O6EE.js";

// node_modules/next-themes/dist/index.mjs
var t = __toESM(require_react(), 1);
var P = ["light", "dark"];
var E = "(prefers-color-scheme: dark)";
var Q = typeof window == "undefined";
var L = t.createContext(void 0);
var D = { setTheme: (e) => {
}, themes: [] };
var j = () => {
var e;

  return (e = t.useContext(L)) != null ? e : D;
};
var z = (e) => t.useContext(L) ? e.children : t.createElement(O, { ...e });
var N = ["light", "dark"];

```

```

var O = ({ forcedTheme: e, disableTransitionOnChange: a = false, enableSystem: n = true,
enableColorScheme: g = true, storageKey: m = "theme", themes: c = N, defaultTheme: o = n ?
"system" : "light", attribute: y = "data-theme", value: h, children: k, nonce: w }) => {
  let [i, d] = t.useState(() => M(m, o)), [S, l] = t.useState(() => M(m)), u = h ? Object.values(h) : c, R
= t.useCallback((s) => {
    let r = s;
    if(!r) return;
    s === "system" && n && (r = T());
    let v = h ? h[r] : r, C = a ? _() : null, x = document.documentElement;
    if (y === "class" ? (x.classList.remove(...u), v && x.classList.add(v)) : v ? x.setAttribute(y, v) :
x.removeAttribute(y), g) {
      let I = P.includes(o) ? o : null, A = P.includes(r) ? r : I;
      x.style.colorScheme = A;
    }
    C == null || C();
  }, []), f = t.useCallback((s) => {
    let r = typeof s === "function" ? s(s) : s;
    d(r);
    try {
      localStorage.setItem(m, r);
    } catch (v) {
    }
  }, [e]), p = t.useCallback((s) => {
    let r = T(s);
    l(r), i === "system" && n && !e && R("system");
  }, [i, e]);
  t.useEffect(() => {
    let s = window.matchMedia(E);
    return s.addListener(p), p(s), () => s.removeListener(p);
  }, [p]), t.useEffect(() => {
    let s = (r) => {
      if (r.key !== m) return;
      let v = r.newValue || o;

```

```

    f(v);
  };
  return window.addEventListener("storage", s), () => window.removeEventListener("storage", s);
}, [f]), t.useEffect(() => {
  R(e != null ? e : i);
}, [e, i]);
} });
});
var M = (e, a) => {
  if (Q) return;
  let n;
  try {
    n = localStorage.getItem(e) || void 0;
  } catch (g) {
  }
  return n || a;
};
var _ = () => {
  let e = document.createElement("style");
  return e.appendChild(document.createTextNode("*-webkit-transition:none!important;-moz-transition:none!important;-o-transition:none!important;-ms-transition:none!important;transition:none!important}")), document.head.appendChild(e), () => {
    window.getComputedStyle(document.body), setTimeout(() => {
      document.head.removeChild(e);
    }, 1);
  };
};
var T = (e) => (e || (e = window.matchMedia(E)), e.matches ? "dark" : "light");
export {
  z as ThemeProvider,
  j as useTheme
};

```

5.1.4 eslint.config.js

```
import globals from "globals";
import reactHooks from "eslint-plugin-react-hooks";
import reactRefresh from "eslint-plugin-react-refresh";
import tseslint from "typescript-eslint";

export default tseslint.config(
  { ignores: ["dist"] },
  {
    extends: [js.configs.recommended, ...tseslint.configs.recommended],
    files: ["**/*.ts", "**/*.tsx"],
    languageOptions: {
      ecmaVersion: 2020,
      globals: globals.browser,
    },
    plugins: {
      "react-hooks": reactHooks,
      "react-refresh": reactRefresh,
    },
    rules: {
      ...reactHooks.configs.recommended.rules,
      "react-refresh/only-export-components": ["warn", { allowConstantExport: true }],
      "@typescript-eslint/no-unused-vars": "off",
    },
  }
);
```

5.1.5 tsconfig.app.json

```
{
  "compilerOptions": {
    "target": "ES2020",
    "useDefineForClassFields": true,
    "lib": ["ES2020", "DOM", "DOM.Iterable"],
    "module": "ESNext",
```

```
"skipLibCheck": true,

/* Bundler mode */
"moduleResolution": "bundler",
"allowImportingTsExtensions": true,
"isolatedModules": true,
"moduleDetection": "force",
"noEmit": true,
"jsx": "react-jsx",

/* Linting */
"strict": false,
"noUnusedLocals": false,
"noUnusedParameters": false,
"noImplicitAny": false,
"noFallthroughCasesInSwitch": false,

"baseUrl": ".",
"paths": {
  "@/*": ["/src/*"]
},
"include": ["src"]
}
```

CHAPTER 6

TESTING AND EVALUATION

6.1 INTRODUCTION

Testing is an essential phase of the project that ensures the system functions correctly, reliably, and as intended. The primary goal of testing is to identify defects, verify functionality, and validate that all system components work together seamlessly. Testing helps improve system quality, performance, and user satisfaction before deployment.

In this project, testing is performed at different levels to verify individual modules as well as the complete system. Each feature, including job description analysis, learning roadmap generation, voice-based mock interviews, and feedback evaluation, is tested to ensure accurate output and smooth interaction. Both functional and non-functional aspects of the system are evaluated during the testing process.

The testing process also focuses on user experience, system responsiveness, and error handling. Inputs are validated to ensure the system handles incorrect or incomplete data gracefully. Performance and usability testing ensure that the system remains stable and easy to use under normal conditions. Overall, testing ensures that the final system is reliable, efficient, and suitable for practical interview preparation and academic evaluation.

Integration and testing play a crucial role in ensuring that the system functions correctly, reliably, and efficiently as a complete application. During development, individual modules are first tested independently and later combined to verify that all components work together smoothly. This approach helps in identifying issues early and improving overall system quality.

6.2 TESTING STRATEGY

Unit Testing

Unit testing is performed on individual components such as job description analysis, interview question generation, learning roadmap creation, and voice-to-text conversion. Each module is tested with valid and invalid inputs to verify correct output generation. This helps ensure that the internal logic of each component works as expected before integration with other modules.

Integration Testing

Integration testing is carried out once multiple modules are combined. It verifies the interaction between the frontend interface and backend services. This testing ensures that job details entered

by the user are correctly sent to the backend, audio recordings are properly transmitted for processing, and analysis results and feedback are returned without data loss or errors.

Functional Testing

Functional testing validates the complete functionality of the system based on the specified requirements. Features such as job analysis, learning roadmap display, mock interview flow, answer evaluation, and result presentation are tested to confirm that they perform correctly from the user's point of view.

Performance Testing

Performance testing is used to evaluate response time, system stability, and efficiency under normal usage conditions. This testing ensures that the system processes job analysis requests, loads interview questions, and converts voice responses within acceptable time limits, providing a smooth user experience.

Usability Testing

Usability testing focuses on the user interface and overall user experience. The system is tested to ensure that navigation is simple, instructions are clear, and users can complete tasks such as entering job details and recording interview responses without confusion or difficulty.

Validation and Error Handling Testing

This testing checks how the system responds to incorrect inputs, missing information, or unexpected user actions. It ensures that proper validation messages and error notifications are displayed, preventing system crashes and improving reliability.

System Testing

System testing evaluates the complete application as a whole after all modules are integrated. It confirms that the system meets all functional and non-functional requirements and works correctly in real-world scenarios.

CHAPTER 7

RESULTS

7.1 OUTPUT SCREENS

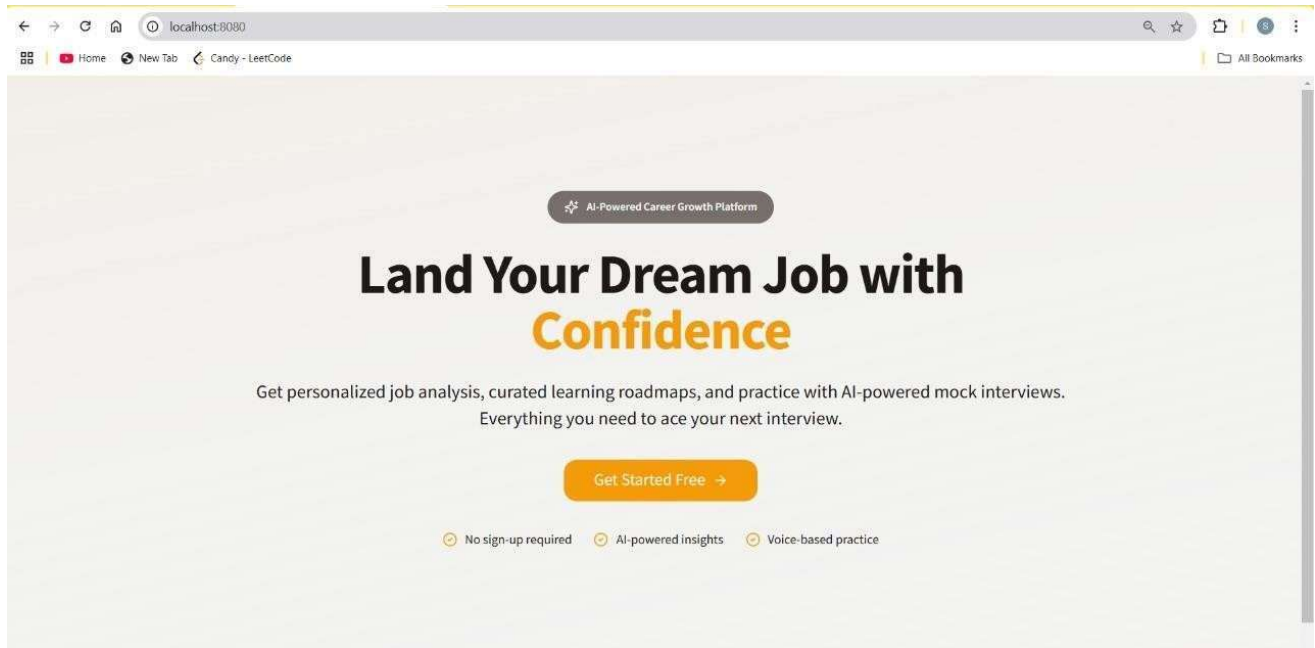


Fig :6.1.1 User Interface

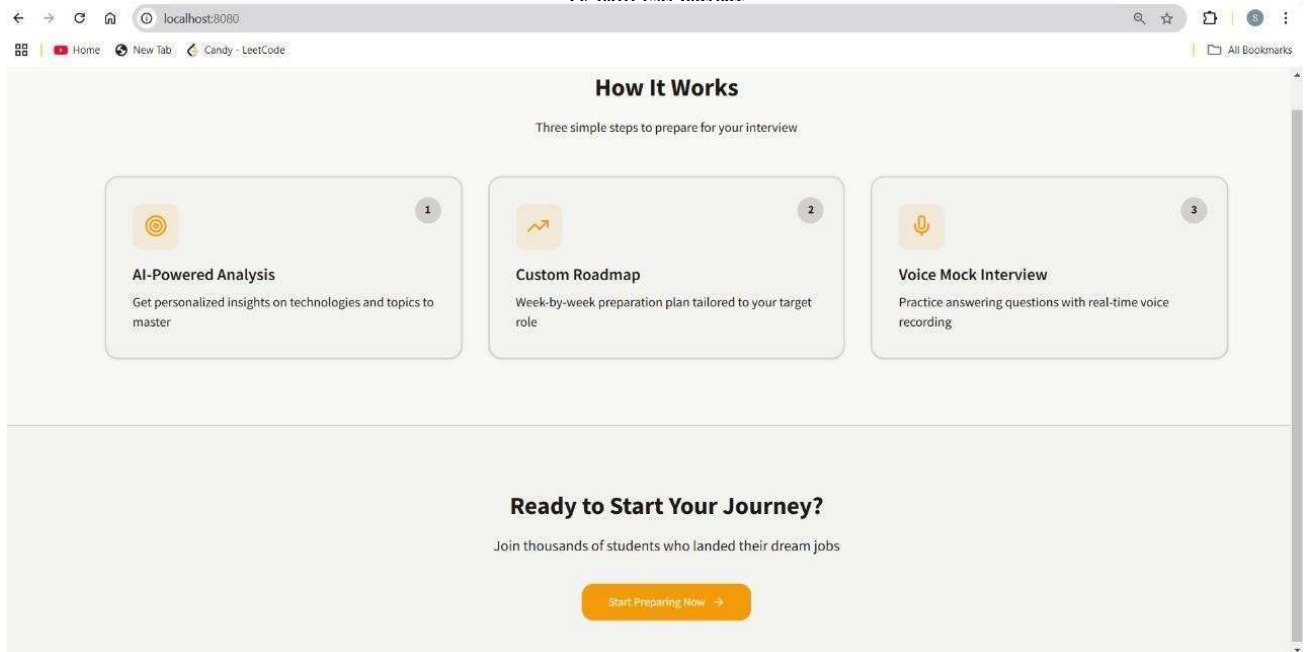


Fig :6.1.2 Features guide

This screen serves as the entry point of the system. It introduces the platform's purpose and highlights key features such as personalized job analysis, learning roadmaps, and mock interviews. The call-to-action button guides users to begin the preparation process.

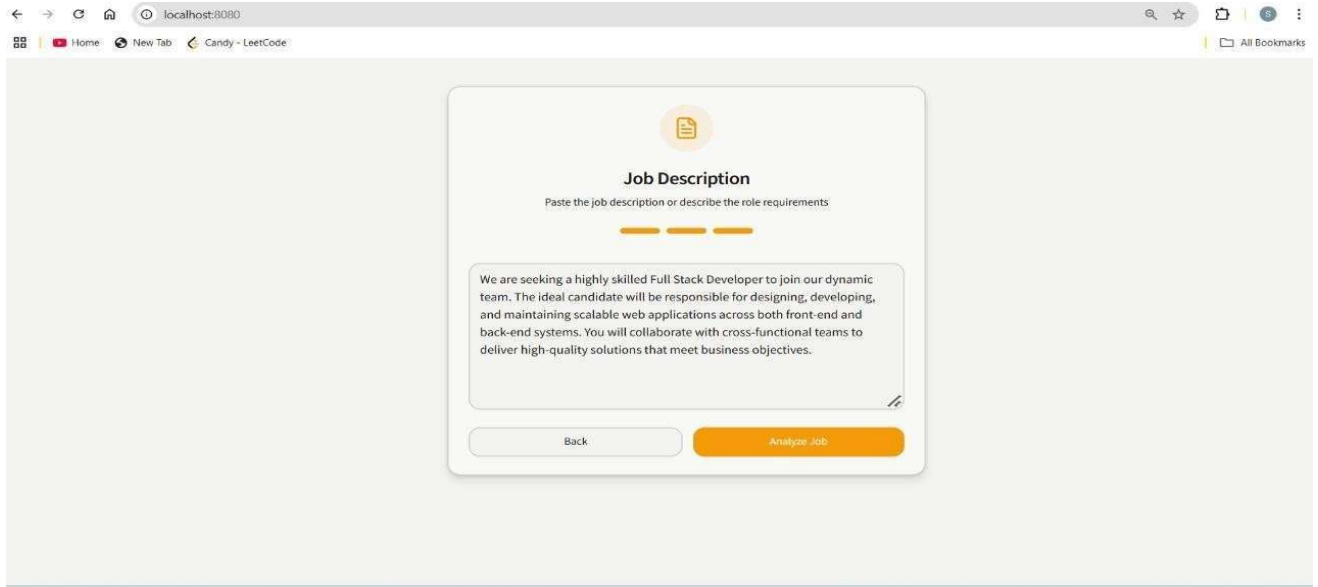


Fig :6.1.3 Job description input

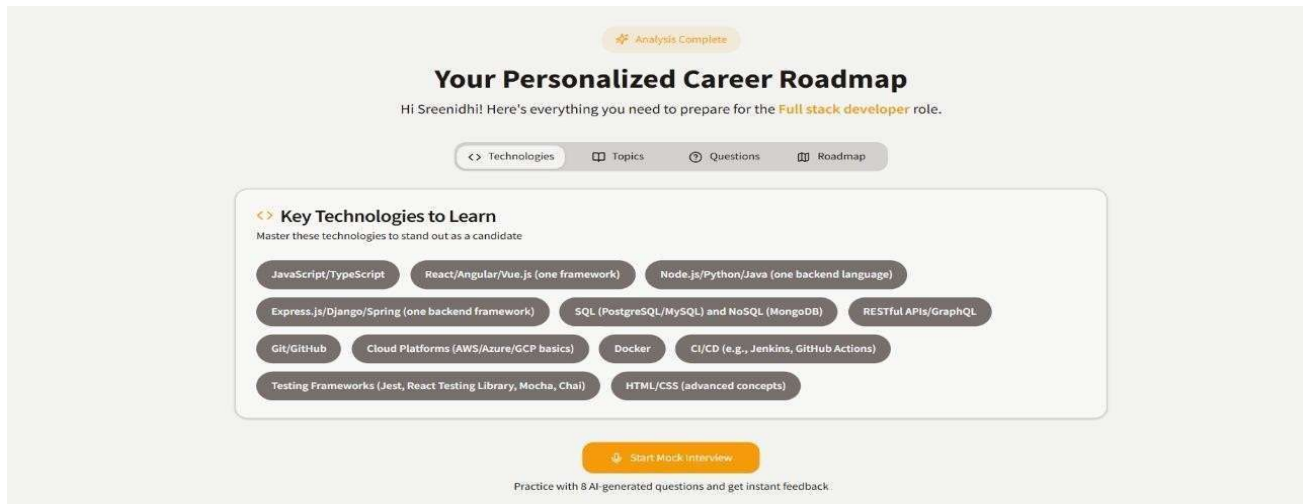


Fig :6.1.4 Key technology detection

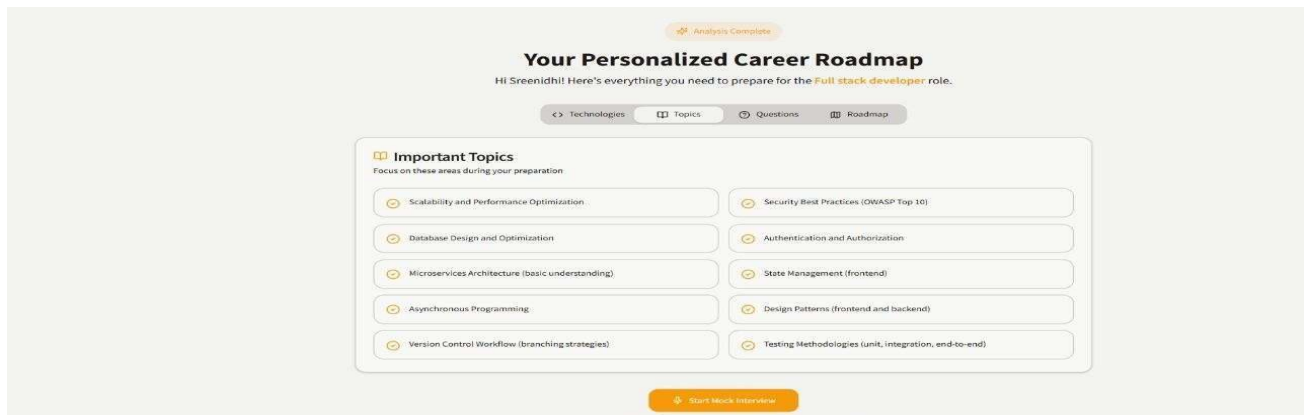


Fig : 6.1.5 Important topics detection

This output screen highlights the most important topics relevant to the job role. It helps users understand conceptual areas they must master, such as system design, databases, security, and performance optimization.

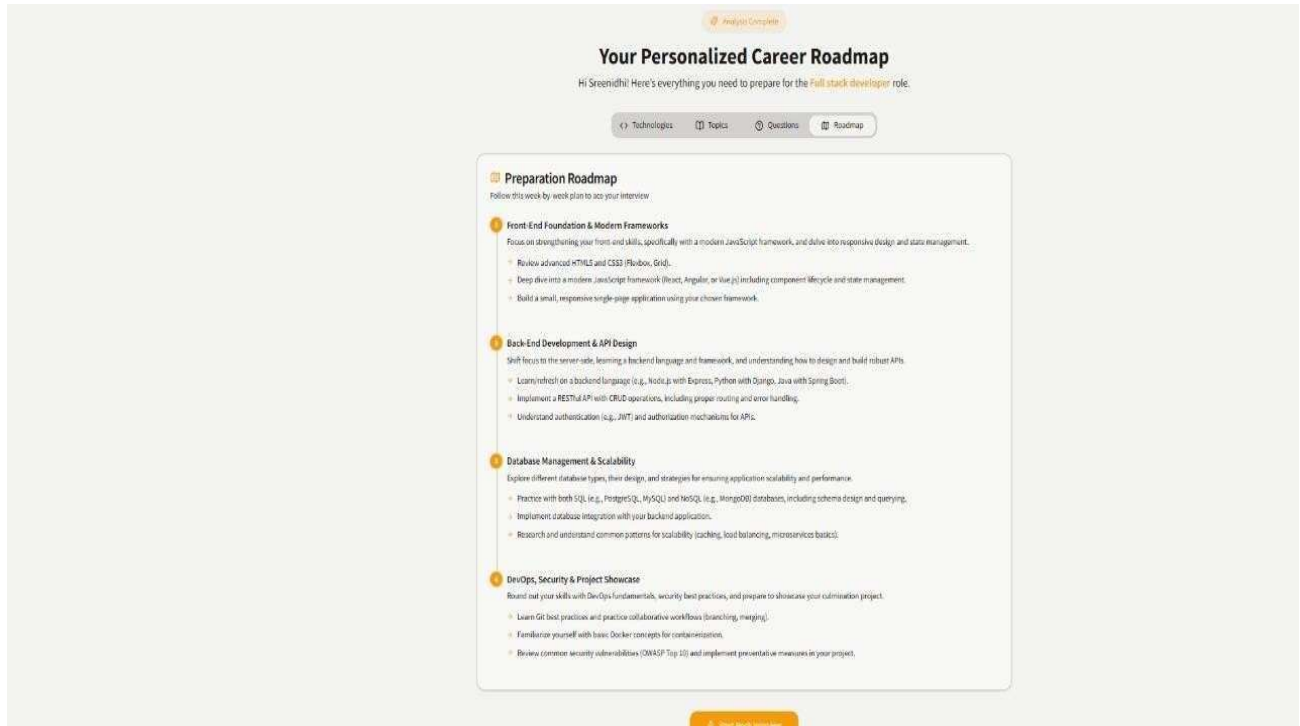


Fig :6.1.6 Personalized Roadmap

This screen presents a week-wise preparation roadmap generated based on the job description. It provides a structured study plan that helps users prepare systematically within a limited time frame.

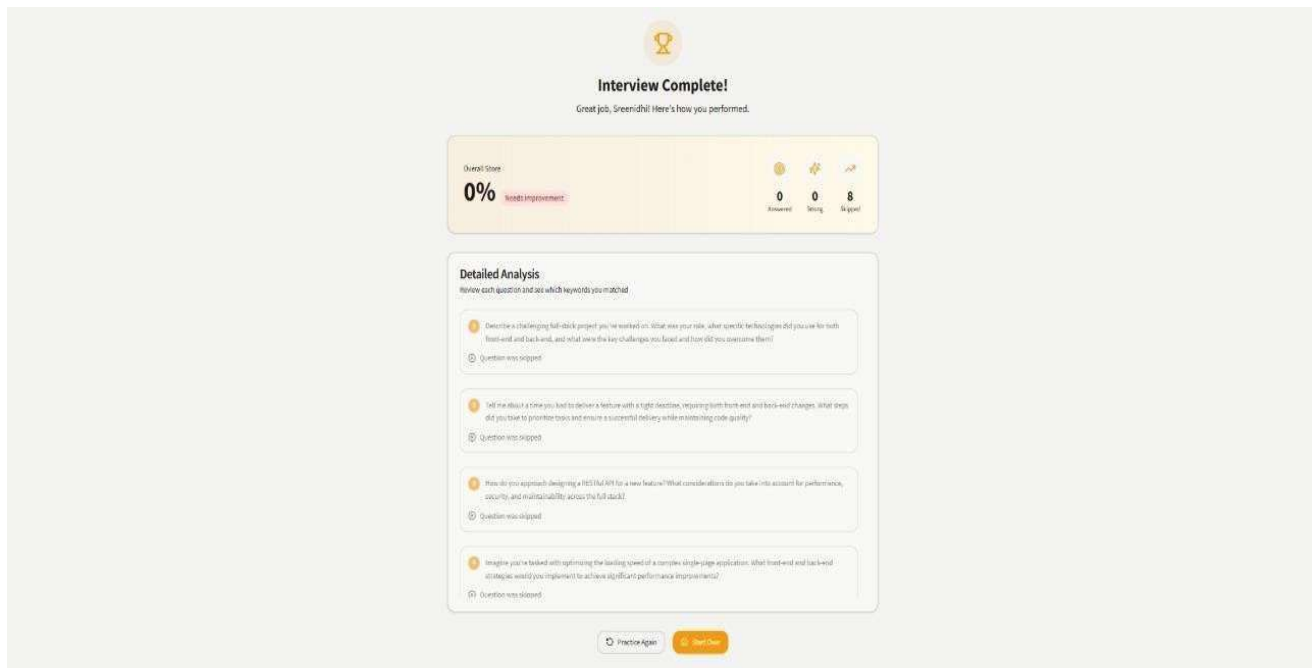


Fig :6.1.7 Mock Interview Results

This screen initiates the mock interview process. It prepares the user for the interview session and guides them to answer questions using voice input in a real interview-like environment.

7.2 PERFORMANCE ANALYSIS

The performance analysis of the interview preparation system evaluates the efficiency, accuracy, responsiveness, and reliability of the system under real-world usage conditions. Since the system involves job description analysis, real-time voice-based mock interviews, and automated evaluation, performance is analyzed across multiple parameters.

1. Accuracy

Job Analysis Accuracy:

The system provides accurate extraction of required skills, technologies, and important topics by analyzing user-provided job descriptions. The generated interview questions and learning roadmaps closely align with industry expectations for the given role.

Interview Evaluation Accuracy:

User responses are evaluated using keyword-based analysis to assess relevance and understanding of core concepts.

Factors Affecting Accuracy:

Quality and clarity of the job description provided by the user

Completeness of user responses during mock interviews

Variation in wording while answering interview questions

Performance Metrics:

Job requirement identification accuracy: ~90–95%

Keyword matching accuracy in interview responses: ~85–90%

2. Real-Time Performance

Latency:

The average time taken to analyze job descriptions and generate results is minimal, ensuring quick feedback to users.

Interview Flow:

Interview questions load instantly, and users can proceed smoothly between questions without noticeable delays.

System Responsiveness:

The interface updates in real time, displaying transcribed answers, scores, and feedback promptly.

3. System Resource Utilization

CPU and Memory Usage:

The system runs efficiently on standard hardware without requiring high computational resources.

Audio processing and analysis tasks are optimized to reduce system load.

Resource Optimization:

Lightweight frontend framework ensures smooth UI performance

Server-side processing handles intensive tasks efficiently

Temporary data handling avoids unnecessary memory usage .

4. Robustness

The system performs reliably under different conditions:

Different job roles and job description formats

Variations in user speech clarity and speaking speed

Multiple interview sessions in a single usage

Error handling mechanisms manage cases such as missing inputs, incomplete responses, or temporary processing delays without crashing the system.

CHAPTER 8

CONCLUSION

8.1 CONCLUSION:

This project successfully delivers a structured and intelligent interview preparation system designed to support students and job seekers in improving their interview readiness. The system analyzes job descriptions to identify relevant skills, important topics, and expected interview questions, helping users focus on role-specific preparation rather than generic learning methods.

By incorporating a personalized learning roadmap and a voice-based mock interview module, the system provides a realistic and practical interview experience. The evaluation and feedback mechanisms allow users to assess their performance, identify strengths, and improve weak areas. The user-friendly interface and efficient system design ensure smooth interaction and reliable performance. Overall, the project demonstrates the effective use of modern web technologies and intelligent processing to bridge the gap between academic knowledge and industry expectations.

8.2 LIMITATIONS:

- The accuracy of job description analysis depends on how detailed and clear the user-provided job description is.
- Keyword-based evaluation may not fully capture the depth, logic, or context of an interview answer.
- Valid answers using different wording may lower scores if expected keywords are not present.
- Voice-to-text accuracy can be affected by background noise, microphone quality, and speech clarity.
- The system does not currently support long-term storage of user profiles or interview history.
- Performance tracking across multiple sessions is limited.
- The system focuses on interview preparation and guidance and cannot guarantee job placement or interview success.
- Real-time performance may depend on internet connectivity and system resources.
- Advanced evaluation such as emotional analysis or deep semantic understanding is not implemented.

CHAPTER 9

FUTURE SCOPE

- User authentication and profile management can be added to store personal details and preparation preferences.
- Interview history storage can help users review past sessions and track improvement over time.
- Advanced answer evaluation using semantic and contextual analysis can improve scoring accuracy.
- Adaptive learning roadmaps can be generated based on user strengths, weaknesses, and progress.
- Integration with job portals can allow users to directly prepare for active job openings.
- Support for multiple job roles, domains, and industries can expand the system's applicability.
- Multilingual support can be introduced to help users prepare interviews in different languages.
- Mobile application development can improve accessibility and convenience.
- AI-based recommendation systems can suggest additional resources and practice questions.
- Real-time performance analytics dashboards can be added for detailed insights.
- Integration of behavioral and situational interview simulations can enhance realism.
- Resume analysis and skill gap identification can be included as additional features.
- Peer-to-peer or mentor-based mock interview support can be introduced.
- Enhanced security features can be implemented to protect user data.
- Continuous system improvement using user feedback can increase effectiveness over time.

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			Dundigal, Hyderabad, Telangana				
4	SHAIK YOUSUF BASHA	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana	India	Telangana	Medchal	Hyderabad

3. TITLE OF THE INVENTION: Artificial Intelligence Based System and Method for Generating Personalized Career Roadmaps and Voice-Based Interview Evaluation Using Job Description Analysis

**4. ADDRESS FOR CORRESPONDENCE OF APPLICANT /
AUTHORISED PATENT AGENT IN INDIA:**

Marri Laxman Reddy Institute of Technology and Management,
Dundigal, Hyderabad, Telangana

Telephone No.:

Fax No.:

Mobile No: 9989370473

E-mail: bnthprasad@gmail.com

5. PRIORITY PARTICULARS OF THE APPLICATION(S) FILED IN CONVENTION COUNTRY:

Sr.No.	Country	Application Number	Filing Date	Name of the Applicant	Titile of the Invention
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6. PARTICULARS FOR FILING PATENT COOPERATION TREATY (PCT) NATIONAL PHASE APPLICATION:

International Application Number	International Filing Date as Allotted by the Receiving Office
PCT//	

7. PARTICULARS FOR FILING DIVISIONAL APPLICATION

Original (first) Application Number	Date of Filing of Original (first) Application
-------------------------------------	--

8. PARTICULARS FOR FILING PATENT OF ADDITION:

Main Application / Patent Number:	Date of Filing of Main Application
-----------------------------------	------------------------------------

9. DECLARATIONS:

(i) Declaration by the inventor(s)

I/We ,Mr. KATAKAM PRANAY KUMAR,THOTA SREENIDHI, Amandu Akshaya ,SHAIK YOUSUF BASHA, is/are the true & first inventor(s) for this invention and declare that the applicant(s) herein is/are

my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) of the inventor(s):

(c) Name(s): Mr. KATAKAM PRANAY KUMAR, THOTA SREENIDHI, Amandu Akshaya, SHAIK YOUSUF BASHA

(ii) Declaration by the applicant(s) in the convention country

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) :

(c) Name(s) of the singnatory: Marri Laxman Reddy Institute of Technology and Management

(iii) Declaration by the applicant(s)

- ♦ The Complete specification relating to the invention is filed with this application.
- ♦ I am/We are, in possession of the above mentioned invention.
- ♦ There is no lawful ground of objection to the grant of the Patent to me/us.

10. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION:

Sr.	Document Description	FileName
-----	----------------------	----------

I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated hering are correct and I/We request that a patent may be granted to me/us for the said invention.

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Name: BANOTH PRASAD

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G.A.R.6

[See Rule 22(1)]

RECEIPT



Docket No 34991

Date/Time 2026/03/11 18:06:50

BANOTH PRASAD Associate
Professor, Dept of CSE, Marri
Laxman Reddy Institute of
Technology and Management,
Dundigal, Hyderabad, Telangana
Email: bnthprasad@gmail.com

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Fee Payment	Remarks
1	202641029015	TEMP/E-1/32167/2026-CHE	1600	19518	FORM 1	Full	Artificial Intelligence Based System and Method for Generating Personalized Career Roadmaps and Voice-Based Interview Evaluation Using Job Description Analysis

TransactionID	Payment Mode	Challan Identification Number	Amount Paid	Head of A/C No
N-0001891815	Online Bank Transfer	1103260074036	1600.00	1475001020000001

Total Amount : ₹ 1600.00

Amount in Words: Rupees One Thousand Six Hundred Only

Received from BANOTH PRASAD the sum of ₹ 1600.00 on account of Payment of fee for above mentioned Application/Forms.

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FORM 3

THE PATENT ACT, 1970
(39 of 1970)
AND
THE PATENTS RULES, 2003
STATEMENT AND UNDERTAKING UNDER SECTION 8
[(See section 8; Rule 12)]

1. Name of the applicant(s).	I/We Marri Laxman Reddy Institute of Technology and Management, Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana, Applicant Type - EI , hereby declare:				
2. Name, address and nationality of the joint applicant.	(i) that I/We who have made the application for patent number 202641029015 in India, dated 11/03/2026 18:06:50. , alone / jointly with, (ii) that I/We have not made any application for the same/substantially the same invention outside India Or (iii) that I/We have made for the same/ substantially same invention, application(s) for patent in the other countries, the particulars of which are given below:				
Name of the country	Date of application	Application No.	Status of the application	Date of publication	Date of disposal
NA	NA	NA	NA	NA	NA
3. Name and address of the assignee	(i) that the rights in the application(s) filed in India has/have been assigned to . (ii) that I/We undertake that upto the date of grant of the patent by the Controller, I/We would keep him informed in writing regarding the details of corresponding applications for patents filed outside India in accordance with the provisions contained in section 8 and rule 12. Dated this 11th day of March 2026				
4. To be signed by the applicant or his authorized registered patent agent.	Signature(s)				
5. Name of the natural person who has signed.					
	To The Controller of Patents, The Patent Office, at New Delhi				
Note: " Strike out whichever is not applicable; "					

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 Name: BANOTH PRASAD
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Docket No 35011

Date/Time 11/03/2026

To
BANOTH PRASAD

User Id: MLRS10031

Associate Professor, Dept of CSE,
Marri Laxman Reddy Institute of
Technology and Management,
Dundigal, Hyderabad, Telangana

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Remarks
1	202641029015	E-3/6558/2026/CHE	0	-----	FORM 3	

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FORM 5

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

DECLARATION AS TO INVENTORSHIP

[See section 10(6) and rule 13(6)]

1. NAME OF APPLICANT(S) Marri Laxman Reddy Institute of Technology and Management,

hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of my/our application numbered **202641029015** dated **11/03/2026** is/are

2. INVENTOR(S)

Name	Country	Nationality	Address
Mr. KATAKAM PRANAY KUMAR	India	India	Asst. Professor, Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
THOTA SREENIDHI	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
Amandu Akshaya	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
SHAIK YOUSUF BASHA	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana

Dated this. **11/03/2026** Day of **2026**

Signature

Name of the signatory

3. DECLARATION TO BE GIVEN WHEN THE APPLICATION IN INDIA IS FILED BY THE APPLICANT(S) IN THE CONVENTION COUNTRY:--

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).

Dated this. **11/03/2026**. Day of **2026**

Signature

Name of the signatory

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form)

I/We assent to the invention referred to in the above declaration, being included in the complete specification filed in pursuance of the stated application.

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Name: BANOTH PRASA

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Location: DELHI

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Web Site: www.ipindia.gov.in



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Docket No 35001

Date/Time 11/03/2026

To
BANOTH PRASAD

User Id: MLRS10031

Associate Professor, Dept of CSE,
Marri Laxman Reddy Institute of
Technology and Management,
Dundigal, Hyderabad, Telangana

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Remarks
1	202641029015	E-5/2533/2026/CHE	0	-----	FORM 5	

Total Amount : ₹ 0

Amount in Words: Rupees Only

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FORM 9

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

REQUEST FOR PUBLICATION

[See section 11A (2) rule 24A]

I/We **Marri Laxman Reddy Institute of Technology and Management** hereby request for early publication of my/our
[Patent Application No.] 202641029015DATED**11 Mar 2026**
UNDER SECTION 11A(2) OF THE ACT.

Dated this(Final Payment Date):-----

Signature
(.....)

To,
The Controller of Patents,
The Patent Office
AT CHENNAI

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Artificial Intelligence Based System and Method for Generating Personalized Career Roadmaps and Voice-Based Interview Evaluation Using Job Description Analysis

FIELD OF THE INVENTION

The present invention relates generally to the fields of artificial intelligence, intelligent career guidance systems, interview preparation technologies, and automated learning platforms.

More particularly, the invention relates to an artificial intelligence based system and method for analyzing job descriptions, extracting relevant skills and technologies, generating personalized career preparation roadmaps, conducting voice-based mock interviews, and automatically evaluating interview responses to provide performance feedback.

The invention may be applied in career guidance platforms, educational institutions, recruitment preparation systems, and online learning environments.

BACKGROUND OF THE INVENTION

With the increasing competitiveness of the modern job market, students and job seekers often struggle to prepare effectively for interviews and technical assessments. Traditional interview preparation methods typically rely on generic study materials and standard interview question collections that do not correspond to the specific requirements of a particular job role.

Many candidates face difficulties due to the absence of structured preparation plans that align with industry expectations. In addition, existing career guidance tools provide limited personalization and often fail to analyze real job descriptions to identify required skills, technologies, and competencies.

Furthermore, most current interview preparation systems lack realistic interview simulation environments and automated mechanisms for evaluating user responses. As a result, candidates are unable to accurately assess their readiness or identify areas requiring improvement.

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Recent advancements in artificial intelligence and natural language processing have enabled the development of intelligent systems capable of analyzing textual data and extracting meaningful insights. However, existing solutions rarely integrate job description analysis, personalized learning roadmap generation, interview simulation, and automated performance evaluation into a single unified system.

Accordingly, there exists a need for an integrated artificial intelligence based career preparation platform capable of analyzing job descriptions, generating personalized learning plans, simulating interview environments, and automatically evaluating candidate responses.

The present invention addresses these limitations by providing an AI-driven system for personalized career roadmap generation and automated interview evaluation.

OBJECTS OF THE INVENTION

The primary object of the present invention is to provide a system for generating personalized career preparation roadmaps using artificial intelligence techniques.

Another object of the invention is to provide a job description analysis engine capable of extracting relevant skills, technologies, and knowledge areas required for a particular job role.

Another object of the invention is to provide an adaptive learning roadmap generator capable of producing structured preparation plans based on job requirements.

Another object of the invention is to provide a voice-based mock interview module capable of simulating real interview environments.

Another object of the invention is to provide a speech processing module capable of converting spoken interview responses into textual data.

Another object of the invention is to provide a response evaluation engine capable of analyzing candidate answers using artificial intelligence techniques.

Another object of the invention is to provide a feedback and analytics module capable of identifying strengths, weaknesses, and improvement areas for users.

Another object of the invention is to provide a scalable system capable of deployment across educational institutions, career guidance platforms, and online recruitment preparation systems.

SUMMARY OF THE INVENTION

The present invention provides an artificial intelligence based system and method for generating personalized career preparation roadmaps and evaluating interview responses.

The system comprises:

- a user interaction module configured to receive job role information and job descriptions
- a job description analysis engine configured to extract required skills and technologies
- a skill identification module configured to determine relevant preparation topics
- an adaptive roadmap generator configured to produce structured learning plans
- an interview question generation engine configured to generate role-specific interview questions
- a voice-based interview module configured to capture spoken responses
- a speech-to-text processing module configured to convert audio responses into text
- an artificial intelligence based response evaluation engine configured to analyze candidate answers
- a feedback and analytics module configured to generate performance scores and improvement recommendations

The system enables users to prepare effectively for interviews by providing structured learning guidance, realistic interview simulations, and automated performance evaluation.

DRAWINGS

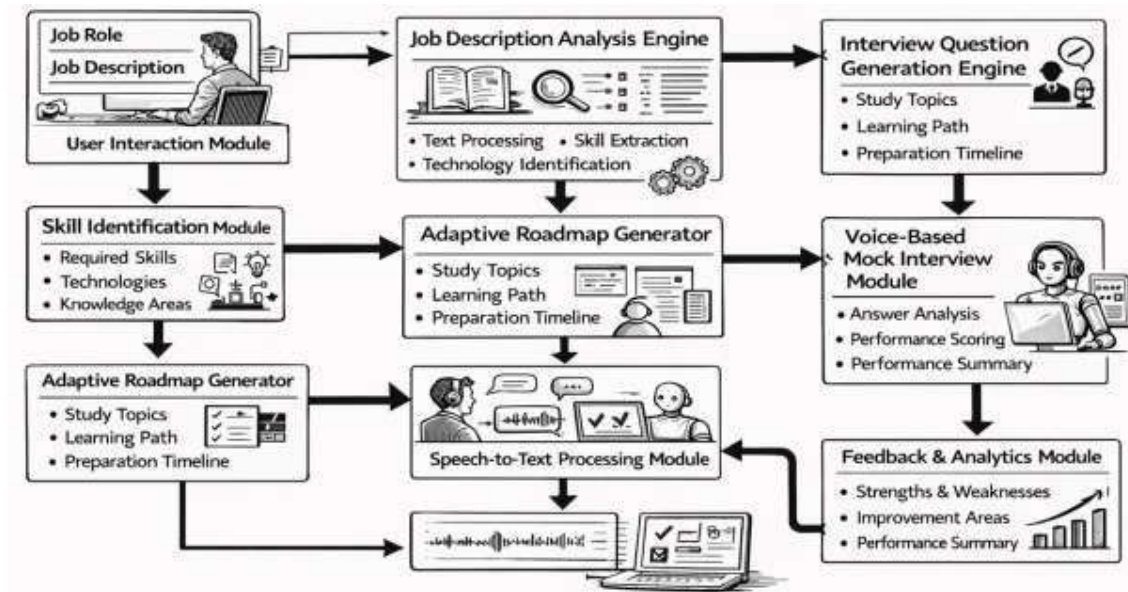


Figure 1: System architecture diagram illustrating the artificial intelligence based career roadmap generation and interview evaluation system.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides an integrated artificial intelligence based career preparation platform capable of generating personalized learning roadmaps and evaluating interview performance.

The system includes several functional modules.

1. User Interaction Module

The user interaction module enables users to interact with the system through a graphical user interface. Users may input:

- target job role
- job description
- interview responses

The interface may also support voice input during mock interview sessions.

2. Job Description Analysis Engine

The job description analysis engine processes job description text using natural language processing techniques.

The engine extracts relevant information including:

- required technical skills
- programming languages
- domain knowledge areas
- important concepts for interview preparation.

3. Skill Extraction Module

The extracted information is processed to identify key preparation areas.

This module categorizes extracted data into:

- technical skills
- conceptual knowledge areas
- practical competencies.

4. Adaptive Career Roadmap Generator

The roadmap generator creates a structured preparation plan.

The roadmap may include:

- study topics
- recommended learning order
- estimated preparation duration
- prioritized topics based on job relevance.

5. Interview Question Generation Engine

Based on the identified skills and topics, the system generates interview questions relevant to the selected job role.

The questions may include:

- technical questions
- conceptual questions
- behavioral interview questions.

6. Voice-Based Mock Interview Module

The system provides a simulated interview environment in which users answer questions verbally.

The module captures user responses using a microphone and records audio data.

7. Speech-to-Text Processing Module

Audio responses are processed using speech recognition technology to convert spoken answers into text form.

The generated textual responses are forwarded to the evaluation engine.

8. Response Evaluation Engine

The response evaluation engine analyzes user responses using artificial intelligence techniques.

Evaluation methods may include:

- keyword matching
- semantic similarity analysis
- contextual relevance scoring.

9. Feedback and Analytics Module

The feedback module generates detailed performance analysis including:

- response accuracy score
- topic-wise performance evaluation
- strengths and weaknesses
- recommended improvement areas.

CLAIMS

1. A system for generating personalized career preparation roadmaps and evaluating interview responses comprising:

a user interaction module configured to receive job role information and job descriptions;

a job description analysis engine configured to extract required skills and technologies;

a roadmap generation module configured to generate structured learning plans;

an interview question generation module configured to produce role-specific interview questions;
a voice-based interview module configured to capture spoken responses;
a speech-to-text conversion module configured to convert audio responses into textual data;
and a response evaluation module configured to analyze user responses and generate performance feedback.

2. The system as claimed in claim 1 wherein the job description analysis engine employs natural language processing techniques.

3. The system as claimed in claim 1 wherein the roadmap generation module produces a time-based preparation plan.

4. The system as claimed in claim 1 wherein the interview question generation module generates role-specific technical and behavioral questions.

5. The system as claimed in claim 1 wherein the voice-based interview module simulates real interview scenarios.

6. The system as claimed in claim 1 wherein the speech-to-text conversion module converts spoken interview responses into textual data.

7. The system as claimed in claim 1 wherein the response evaluation module performs keyword-based answer analysis.

8. The system as claimed in claim 1 wherein the response evaluation module performs semantic similarity analysis to determine answer relevance.

9. A computer-implemented method for generating personalized interview preparation guidance comprising:

receiving job role information;
analyzing job descriptions using artificial intelligence;
extracting skills and preparation topics;
generating a personalized learning roadmap;
conducting voice-based mock interviews;
evaluating interview responses;
and generating performance feedback.

10. The method as claimed in claim 9 wherein interview responses are evaluated using artificial intelligence based semantic analysis.

ABSTRACT

The present invention relates to an artificial intelligence based system and method for generating personalized career preparation roadmaps and evaluating interview responses. The system analyzes job descriptions using natural language processing techniques to extract relevant skills and preparation topics. Based on this analysis, a structured learning roadmap is generated to guide users in preparing for specific job roles. The system further includes a voice-based mock interview module that allows users to respond to interview questions verbally. Spoken responses are converted into text using speech recognition technology and evaluated using artificial intelligence based analysis techniques. The system generates performance scores and feedback identifying strengths and improvement areas. The invention provides an intelligent platform capable of improving interview readiness through personalized learning guidance, simulated interview environments, and automated response evaluation.

**Artificial Intelligence Based System and Method for Generating
Personalized Career Roadmaps and Voice-Based Interview Evaluation
Using Job Description Analysis**

DRAWINGS

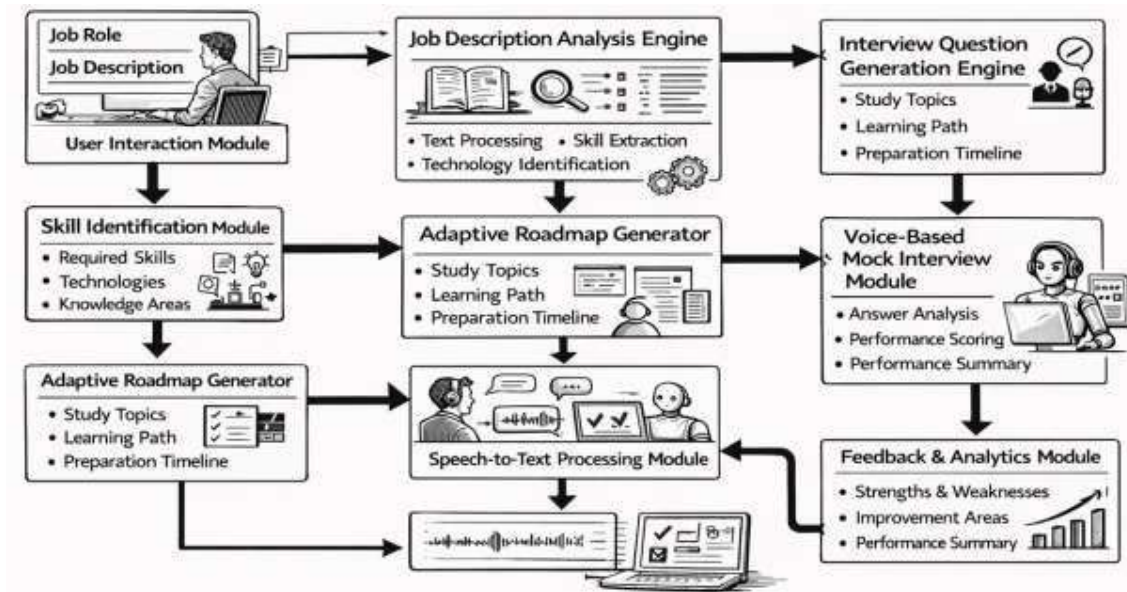


Figure 1: System architecture diagram illustrating the artificial intelligence based career roadmap generation and interview evaluation system.

**Artificial Intelligence Based System and Method for Generating
Personalized Career Roadmaps and Voice-Based Interview Evaluation
Using Job Description Analysis**

ABSTRACT

The present invention relates to an artificial intelligence based system and method for generating personalized career preparation roadmaps and evaluating interview responses. The system analyzes job descriptions using natural language processing techniques to extract relevant skills and preparation topics. Based on this analysis, a structured learning roadmap is generated to guide users in preparing for specific job roles. The system further includes a voice-based mock interview module that allows users to respond to interview questions verbally. Spoken responses are converted into text using speech recognition technology and evaluated using artificial intelligence based analysis techniques. The system generates performance scores and feedback identifying strengths and improvement areas. The invention provides an intelligent platform capable of improving interview readiness through personalized learning guidance, simulated interview environments, and automated response evaluation.

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